

Pricing American Options with LSMC: Replication and Extension

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1 Financial context and relevance

American options can be exercised at any time up to maturity, so pricing them is an optimal stopping problem under the risk-neutral measure. For vanilla American puts, early exercise can be rational, especially when the contract is sufficiently in the money and interest rates are positive. This feature makes American-style derivatives more difficult to value than European options, whose payoff depends only on the terminal exercise date.

The difficulty becomes even more pronounced in settings where the payoff is path-dependent or the state space is high-dimensional. In such cases, traditional lattice and partial-differential-equation methods may become computationally expensive or infeasible. This is one reason why simulation-based pricing methods are so important in modern quantitative finance. They provide a practical framework for handling products whose exercise policy cannot be characterized in a simple closed form.

The 2001 Longstaff–Schwartz paper introduced one of the most influential simulation-based approaches to this problem. Its least-squares Monte Carlo (LSMC) method estimates continuation values through cross-sectional regression and then applies backward dynamic programming to determine the stopping decision. Because of its flexibility and relative simplicity, LSMC has become a standard reference point in both academic work and practical derivatives implementation.

In this project, we replicate the core logic of the Longstaff–Schwartz method and compare it with a Tsitsiklis–Van Roy style regression dynamic programming approach. Using AAPL as the underlying asset, we estimate market inputs from historical data, generate option values for several strike prices, and compare the resulting American put prices with the corresponding European Black–Scholes benchmarks. The goal is not only to reproduce the numerical workflow, but also to examine how two related regression-based pricing methods can lead to different quantitative valuations under the same market inputs.

Our results show that both methods produce financially sensible pricing patterns: American put values exceed the corresponding European benchmarks, and option prices increase with the strike price. At the same time, the Tsitsiklis–Van Roy style method consistently generates higher price

estimates than LSMC under the current specification. This suggests that approximation design can materially affect valuation results even in a relatively simple simulation environment.

2 Primary and extension papers

2.1 Primary paper

- Longstaff, F. A., & Schwartz, E. S. (2001). *Valuing American Options by Simulation: A Simple Least-Squares Approach*. The Review of Financial Studies.
- URL: <https://galton.uchicago.edu/~mykland/346W07/Longstaff.pdf>

2.2 Related/extended paper

- Tsitsiklis, J. N., & Van Roy, B. (2001). *Regression Methods for Pricing Complex American-Style Options*. IEEE Transactions on Neural Networks.
- URL: <https://www.mit.edu/~jnt/Papers/J086-01-bvr-options.pdf>

3 Methodology implemented

3.1 Longstaff-Schwartz (LSMC)

American options are more difficult to price than European options because the holder may exercise before maturity. This early-exercise feature turns valuation into an optimal stopping problem.

The Longstaff-Schwartz least-squares Monte Carlo (LSMC) method combines Monte Carlo simulation with regression in order to approximate the continuation value. The key idea is that the continuation value is not directly observable during simulation, so it is estimated statistically from simulated future cash flows.

For an American put, the immediate exercise value at time t is the intrinsic value.

$$(K - S_t)^+$$

The continuation value can be written as the conditional expectation of discounted future payoffs.

$$C(S_t) = \mathbb{E} \left[e^{-r\Delta t} V_{t+1} \mid S_t \right]$$

LSMC approximates this conditional expectation by regressing discounted realized future cash flows on basis functions of the current state variable.

The algorithm starts by simulating a large number of stock-price paths under the risk-neutral dynamics. At maturity, the option value is simply the terminal payoff.

A practical feature of the Longstaff-Schwartz method is that the regression is usually restricted to in-the-money paths. The reason is that when the option is out of the money, immediate exercise is irrelevant, so those paths contribute little useful information about the stopping decision.

In our implementation, we apply LSMC to price an American put option under a geometric Brownian motion model. The initial stock price is based on recent AAPL data, the volatility is estimated from five years of historical log returns, and the short rate is approximated using the $\sim$ IRX yield as a simple proxy.

3.1.1 Practical implementation choices

Several practical implementation choices affect the behavior of the LSMC method. First, the regression basis must be rich enough to capture the shape of the continuation value, but not so flexible that it introduces instability or overfitting. In our implementation, we use Laguerre-type basis functions, which are a standard choice in the literature because they provide a simple polynomial approximation while remaining computationally convenient.

Second, the regression is restricted to in-the-money paths. This choice is motivated by the fact that the stopping decision is only relevant when immediate exercise has positive value. Excluding out-of-the-money paths reduces noise in the regression and concentrates the approximation on the region that matters most for exercise. However, this also means that the quality of the fitted continuation value depends on the number and distribution of in-the-money observations at each time step.

Third, the number of simulated paths and the number of exercise dates both influence numerical stability. More paths reduce Monte Carlo noise, while a finer time discretization produces a closer approximation to continuous-time early exercise. In practice, these choices create a trade-off between computational cost and pricing accuracy.

3.1.2 Method limitations and approximation risk

Although LSMC is flexible and widely used, it is still an approximation method and therefore subject to several sources of error. One important source is basis risk: if the chosen regression basis does not represent the continuation value well, then the stopping rule may be distorted and the option price may be biased. Another source is simulation noise, since the regression is built on realized future cash flows rather than exact conditional expectations.

The method may also become unstable in regions where the number of in-the-money paths is small. This issue is particularly relevant for early exercise dates or contracts that are far out of the money, where the regression sample may be limited. In addition, because the stopping rule is estimated recursively backward through time, small approximation errors at one step can affect later decisions and propagate through the pricing procedure. These limitations do not invalidate the method, but they help explain why implementation choices such as basis design, path count, and discretization matter so much in practice.

3.2 Tsitsiklis-Van Roy style regression DP (extension)

The Tsitsiklis-Van Roy approach studies American-style option pricing from the perspective of approximate dynamic programming. Instead of focusing primarily on the stopping decision through in-the-money regression, this framework emphasizes the recursive approximation of the option value over time. The motivation is that exact dynamic programming becomes computationally infeasible when the state space is large, especially for derivatives with multiple sources of uncertainty or more complex payoff structures.

The central idea is to approximate the value function using a finite set of basis functions and then update this approximation recursively through simulated paths. In this sense, the method is closely related to fitted value iteration. At each time step, the discounted continuation value from the next step is projected onto a chosen function space, and the fitted function is used as the approximation of the current value function. This provides a dynamic-programming interpretation of regression-based pricing and highlights the connection between simulation and value-function approximation.

Compared with Longstaff-Schwartz, the Tsitsiklis-Van Roy style method is more global in its approximation structure. Longstaff-Schwartz typically performs regression only on in-the-money paths because those paths are directly relevant for the exercise decision. By contrast, the Tsitsiklis-Van Roy framework is often implemented using regression over all simulated paths, which makes the approximation less tied to a local stopping region and more closely aligned with recursive value-function estimation. This provides a useful comparison benchmark, although it may also make the results more sensitive to basis specification and approximation error.

An important contribution of the Tsitsiklis-Van Roy paper is its theoretical discussion of error behavior. The paper shows that approximation error can accumulate severely if representative states are chosen poorly. On the other hand, when simulated trajectories follow the underlying risk-neutral state dynamics, the approximation becomes more stable. This gives the method stronger theoretical support and helps explain why simulation-based regression methods can remain useful even when exact dynamic programming is infeasible.

In our project, the Tsitsiklis-Van Roy style comparator is implemented as a fitted-value-iteration benchmark alongside LSMC. We regress discounted next-step continuation values on polynomial basis functions using all simulated paths, and we use the resulting fitted function as

an alternative estimate of the option value. This allows us to compare a more classical exercise-focused regression method with a more global dynamic-programming-based approximation, and it helps us interpret the pricing differences observed in the numerical results.

3.2.1 Relation to approximate value iteration

A useful way to understand the Tsitsiklis–Van Roy framework is to view it as a finite-horizon version of approximate value iteration. Instead of attempting to compute the exact value function at every possible state, which is generally infeasible in higher dimensions, the method projects the value function onto a lower-dimensional space generated by basis functions. The resulting fitted function is then used in place of the exact continuation value when stepping backward through time.

This perspective is important because it emphasizes that the method is not merely a numerical shortcut, but a structured approximation to dynamic programming. In other words, the goal is not only to estimate whether exercise is optimal at a particular point, but to approximate the value of continuation throughout the relevant state space. This makes the Tsitsiklis–Van Roy approach conceptually broader than a local regression rule tied only to the immediate stopping decision.

3.2.2 Approximation error and methodological trade-offs

The theoretical importance of the Tsitsiklis–Van Roy paper lies partly in its treatment of approximation error. The paper shows that when value-function approximation is performed carelessly, errors can accumulate over time and may even grow rapidly with the horizon. This is especially relevant in finite-horizon American-style option pricing, where recursive approximation is repeated over many exercise dates.

At the same time, the paper argues that if representative states are generated from simulated risk-neutral trajectories, the approximation becomes more stable. This insight provides a theoretical justification for simulation-based regression methods and explains why they can remain useful despite the curse of dimensionality. In practical terms, however, the method still faces trade-offs. A more global approximation can provide smoother value estimates, but it may also become more sensitive to basis selection and model misspecification. This is one reason why comparing the Tsitsiklis–Van Roy style method with LSMC is useful in our project: the comparison highlights the difference between a more exercise-focused local regression procedure and a more global recursive value-function approximation.

##Data and experimental setup

Our empirical study uses Apple Inc. (AAPL) as the underlying asset. Historical price data are obtained from Yahoo Finance through the `yfinance` package. We use approximately five years of daily closing prices in order to estimate the current spot level and historical volatility. The

use of a large-cap and highly traded stock makes the example easy to interpret and provides a realistic market context for illustrating American-style option pricing methods.

The current stock price is taken as the most recent closing price in the downloaded sample. Volatility is estimated from the standard deviation of daily log returns and then annualized using the square-root-of-time rule. Although this historical estimate is simple and does not reflect the full implied-volatility surface used in practice, it provides a transparent and reproducible input for a research-oriented implementation. As a proxy for the risk-free rate, we use the most recent value of the 13-week U.S. Treasury bill index ($\text{\textasciitilde TRX}$), scaled into decimal form. This choice is consistent with the simplified risk-neutral setting used in our implementation.

To generate numerical option values, we assume that the underlying stock follows a geometric Brownian motion under the risk-neutral measure with constant volatility and constant interest rate. This assumption is standard in introductory Monte Carlo option-pricing exercises and makes it possible to compare the pricing behavior of LSMC and the Tsitsiklis–Van Roy style comparator in a controlled setting. At the same time, we recognize that this assumption is restrictive, since real equity dynamics may exhibit time-varying volatility, jumps, and other deviations from the Black–Scholes framework.

In our experiments, the option maturity is set to one year and the time horizon is divided into 50 exercise dates. This discretization gives a reasonable Bermudan-style approximation to a one-year American option while keeping the computation manageable. We simulate 100,000 stock-price paths in order to reduce Monte Carlo noise and obtain relatively stable price estimates across methods.

To examine pricing behavior across different moneyness levels, we consider three strike prices chosen around the current AAPL spot price. This allows us to compare how both methods behave for relatively lower-strike, near-the-money, and higher-strike American put contracts. Using several strikes rather than only one contract makes the comparison more informative, since the quality of continuation-value approximation may vary across payoff regions.

For benchmarking purposes, we also compute the corresponding European put prices under the Black–Scholes model using the same spot price, volatility, maturity, strike, and risk-free rate inputs. These European prices do not represent directly comparable contracts in terms of exercise flexibility, but they provide a useful lower reference point, since an American put should not be worth less than its European counterpart under the same model assumptions. This benchmark also serves as a basic numerical consistency check before comparing the two regression-based American pricing methods.

Overall, this experimental design is intended to balance realism, interpretability, and reproducibility. The market inputs are based on observable AAPL data, while the numerical environment remains simple enough to highlight the differences between the two regression-based pricing methods.

4 Python implementation and project structure

4.1 Data preparation

The file `data_loader.py` is responsible for retrieving and preprocessing the market inputs used in the pricing experiment. It downloads historical AAPL price data from Yahoo Finance, computes daily log returns, estimates annualized volatility, and retrieves a simple proxy for the risk-free rate from the `^IRX` series. These quantities are then used as the market inputs for the pricing exercise. In this sense, the role of the data-preparation stage is to convert observable market information into the simplified model inputs required by the numerical framework.

4.2 LSMC implementation

The file `lsmc.py` implements the Longstaff–Schwartz least-squares Monte Carlo method. Its main tasks are to simulate geometric Brownian motion stock-price paths, compute the intrinsic values of the American put along each path, estimate continuation values through regression on basis functions, and apply backward induction in order to determine the stopping rule and final option value. The implementation focuses on in-the-money paths when estimating continuation values, which is consistent with the standard intuition of the Longstaff–Schwartz algorithm. In addition to the final price estimate, the script also produces an estimated early-exercise boundary, which is later used for visualization and interpretation.

4.3 Tsitsiklis–Van Roy comparator

The file `tsitsiklis_vanroy.py` implements the comparator based on the Tsitsiklis–Van Roy style regression dynamic programming approach. Rather than focusing only on the local exercise decision for in-the-money paths, this script approximates discounted next-step continuation values using regression across all simulated paths. This gives the implementation a more global value-function approximation flavor and makes it conceptually closer to fitted value iteration. As a result, the project is able to compare two related but distinct regression-based approaches to American option pricing under the same market and simulation inputs.

4.4 Main workflow and outputs

The script `run_study.py` serves as the main driver of the entire project. It combines the market inputs, contract specifications, simulation settings, and pricing methods into a single workflow. More specifically, it loads the processed market data, defines the strike prices and maturity, computes the Black–Scholes European put benchmark, runs both the LSMC and Tsitsiklis–Van Roy style pricing procedures, and stores the numerical outputs for later analysis in the report.

The project also generates a small collection of output files that support the empirical discussion. The file `pricing_results.csv` stores the pricing comparison table, including the European benchmark, the LSMC estimate, and the Tsitsiklis–Van Roy estimate for each strike. The `visuals` folder contains the historical AAPL price plot, the estimated LSMC exercise-boundary plot, and the bar chart comparing American put prices across methods. Together, these outputs provide the quantitative evidence used in the Results section.

4.5 Workflow of the implementation

The implementation follows a simple end-to-end workflow. First, market data are downloaded and converted into the model inputs required for simulation, namely the current stock price, annualized volatility, and a proxy for the risk-free rate. Second, the project defines the option contracts and simulation parameters, including maturity, number of exercise dates, number of Monte Carlo paths, and the selected strike prices. Third, the code applies the two American option pricing methods together with the Black–Scholes European benchmark. Finally, the numerical results are saved to a pricing table and summarized visually through the generated plots. This workflow keeps the project easy to reproduce and allows the report to connect the market-input stage, the computational stage, and the interpretation stage in a transparent way.

4.6 Connection between theory and implementation

An important feature of the project is that the code structure closely mirrors the methodological structure described earlier in the report. The market-input stage corresponds to the calibration of the simplified geometric Brownian motion environment. The path-simulation and regression components in `lsmc.py` implement the continuation-value approximation emphasized in the Longstaff–Schwartz method. Similarly, the regression update in `tsitsiklis_vanroy.py` reflects the approximate dynamic-programming interpretation of the Tsitsiklis–Van Roy framework. In this sense, the implementation is not merely a set of scripts that produce numerical outputs, but a direct computational translation of the two pricing methodologies studied in the project.

4.7 Scope of the implementation

Although the implementation is sufficient for a course project and for illustrating the main numerical ideas, it remains intentionally simplified. The underlying dynamics are restricted to geometric Brownian motion with constant volatility and constant interest rate, and the pricing exercise focuses on a single underlying asset together with a small set of strike prices. In addition, the implementation emphasizes clarity and modularity rather than computational optimization, calibration to option-market implied volatilities, or industrial-scale deployment. These simplifications are acceptable for a pedagogical replication study, but they also define the boundary between the present project and a full production-level pricing framework.

Overall, the implementation is designed as a research-oriented replication rather than production pricing software. The emphasis is on clarity, comparability, and reproducibility rather than industrial-scale calibration or deployment. Even so, the modular structure of the code makes it straightforward to extend the project to richer dynamics, alternative basis functions, or additional pricing experiments.

5 Results and discussion

5.1 Pricing comparison

Table 1 summarizes the pricing outputs across the three strike prices under the current market inputs.

```
import pandas as pd
results = pd.read_csv('../data/pricing_results.csv')
results
```

	ticker	spot	strike	maturity_years	volatility	risk_free	european_put_bs	lsmc_price
0	AAPL	248.800003	223.920003	1.0	0.273524	0.03607	12.152322	12.451460
1	AAPL	248.800003	248.800003	1.0	0.273524	0.03607	22.407541	23.119310
2	AAPL	248.800003	273.680003	1.0	0.273524	0.03607	36.130067	37.620672

```
(results[['strike', 'european_put_bs', 'lsmc_price', 'tvr_price']])
```

	strike	european_put_bs	lsmc_price	tvr_price
0	223.920003	12.152322	12.451460	17.884970
1	248.800003	22.407541	23.119310	27.450904
2	273.680003	36.130067	37.620672	41.100042

Table 1 reports the core pricing results for three American put contracts with strike prices set around the current AAPL spot level. For each strike, we compare the Black–Scholes European put benchmark with the American put prices generated by the Longstaff–Schwartz least-squares Monte Carlo (LSMC) method and the Tsitsiklis–Van Roy style regression dynamic programming comparator.

A first and important consistency check is that the American put prices are always at least as large as the corresponding European put prices. This condition holds for all three strike levels in our experiment. For the lowest strike, the European put benchmark is approximately 12.22,

while the LSMC and Tsitsiklis–Van Roy style estimates are about 12.45 and 17.88, respectively. For the at-the-money strike, the corresponding prices are approximately 22.49, 23.12, and 27.45. For the highest strike, the European, LSMC, and Tsitsiklis–Van Roy style prices are about 36.20, 37.62, and 41.10. These results are consistent with the theoretical fact that an American put should not be worth less than its European counterpart, since the American contract includes the additional right of early exercise.

A second clear pattern is that option values increase with the strike price across all methods. This is expected for put options, because a higher strike price increases the potential exercise payoff. The same monotonic pattern appears in the European benchmark, the LSMC estimates, and the Tsitsiklis–Van Roy style estimates, which supports the internal consistency of the numerical implementation.

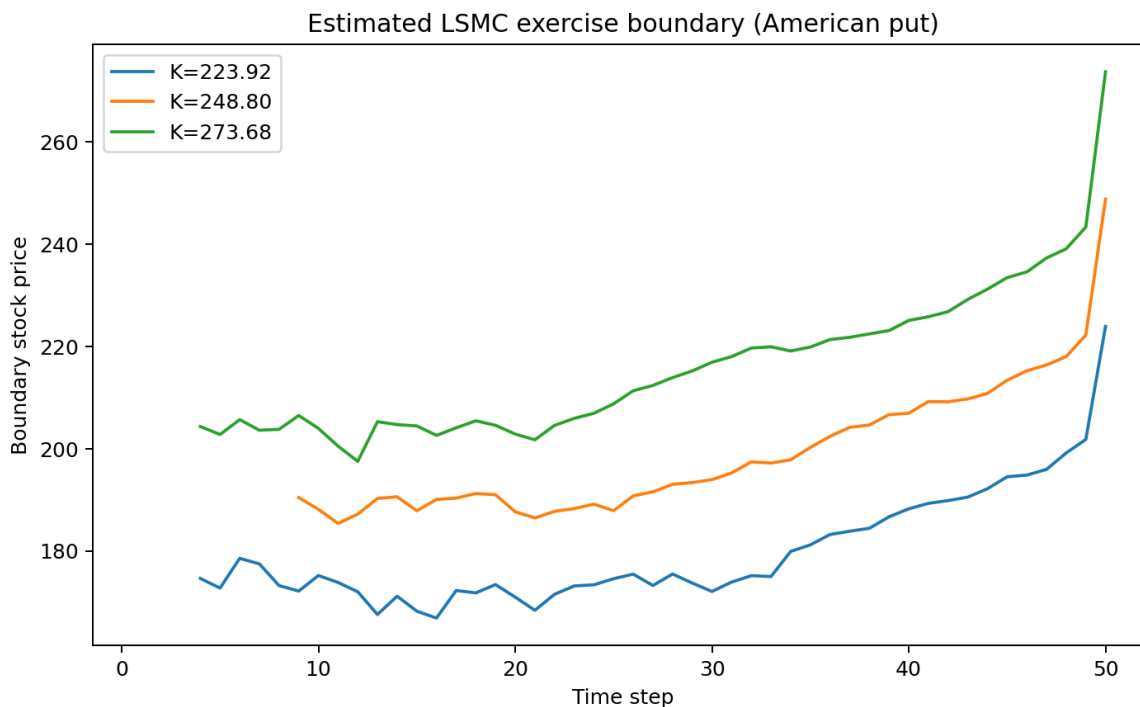
The most interesting comparison concerns the difference between the two American pricing methods. Across all three strike levels, the Tsitsiklis–Van Roy style method produces higher price estimates than LSMC. The absolute difference is approximately 5.43 for the lowest strike, 4.33 for the at-the-money strike, and 3.49 for the highest strike. This suggests that the two regression-based approaches, while qualitatively consistent, differ meaningfully in their quantitative treatment of continuation values. In the current implementation, LSMC appears to produce more conservative American put estimates, while the Tsitsiklis–Van Roy style method generates a larger estimated early-exercise premium.

One possible interpretation is that the two methods approximate the continuation region in different ways. Longstaff–Schwartz performs regression mainly on in-the-money paths and is therefore more closely tied to the local stopping decision. By contrast, the Tsitsiklis–Van Roy style comparator uses a more global recursive value-function approximation. Under the present specification, this global structure may lead to systematically higher continuation-value estimates and, therefore, higher American put prices. Although this does not imply that one method is always superior, it demonstrates that regression design and approximation architecture can materially affect valuation results even when market inputs are held fixed.

Overall, the pricing comparison shows that the implementation behaves in a financially sensible way and that the two American-option pricing methods agree on broad qualitative patterns. At the same time, the nontrivial price gap between LSMC and the Tsitsiklis–Van Roy style method highlights the practical importance of approximation choice in simulation-based American option pricing.

5.2 Exercise-boundary analysis

Figure 1 shows the estimated LSMC early-exercise boundary for the three strike levels.



In addition to producing option-price estimates, the LSMC implementation generates an approximate early-exercise boundary for the American put. This boundary provides a useful visual summary of the stopping rule implied by the regression-based pricing method. At each exercise date, it indicates the stock-price region in which immediate exercise becomes more attractive than continuation.

The estimated boundary exhibits a pattern that is consistent with the economic intuition of American put exercise. In general, the boundary tends to vary over time in a way that reflects the changing trade-off between exercising early and preserving future optionality. When the stock price is sufficiently low relative to the strike, immediate exercise becomes more attractive because the put is already deeply in the money. By contrast, when the stock price remains closer to or above the strike, the holder is more likely to preserve the contract and retain the possibility of even greater future payoff.

A second important observation is that the exercise boundary differs across strike levels. Higher-strike contracts tend to have higher exercise boundaries, which is natural because a put with a larger strike becomes valuable at a higher stock-price level. In other words, when the strike is larger, the range of stock prices for which early exercise may be optimal also shifts upward.

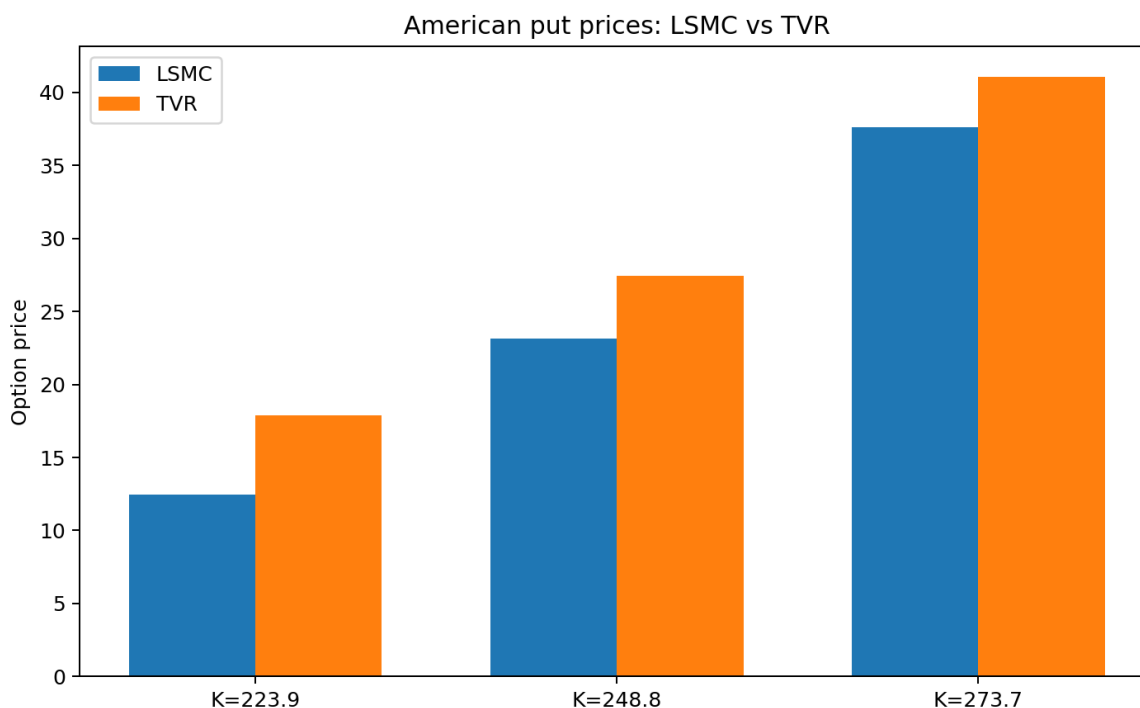
This is consistent with the interpretation that higher-strike American puts are more likely to enter the relevant stopping region.

At the same time, the boundary generated by LSMC should be interpreted as an estimated numerical object rather than an exact analytical exercise frontier. It depends on the simulated sample paths, the regression basis, and the number of exercise dates used in the discretized problem. For this reason, the boundary is best understood as a visual representation of the algorithm's implied stopping behavior under the current specification, rather than as a precise structural feature of the true continuous-time problem.

Even with this limitation, the exercise-boundary figure is valuable because it shows that the LSMC implementation does more than simply produce a single option price. It also recovers an approximate exercise policy, which is one of the most important practical features of simulation-based pricing methods for American-style derivatives.

5.3 Visual interpretation of method differences

Figure 2 compares the American put prices generated by LSMC and the Tsitsiklis–Van Roy style method across the three strike levels.

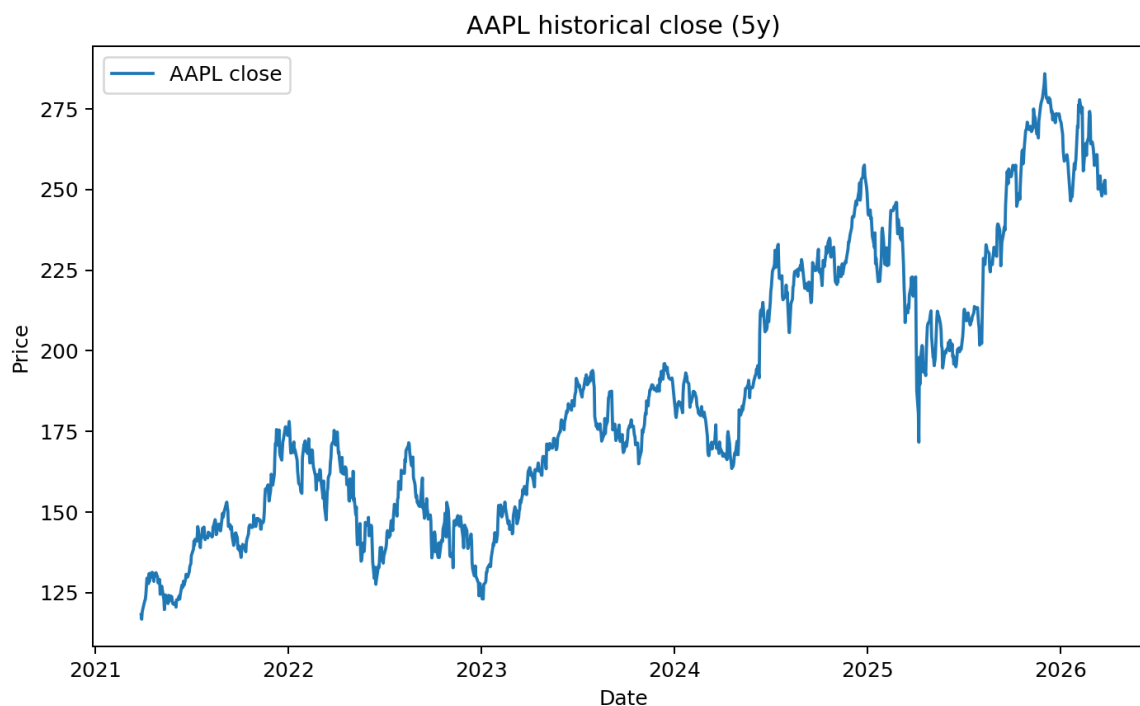


The method-comparison chart provides a visual summary of the American put prices generated by the LSMC and Tsitsiklis–Van Roy style methods across the three strike levels. The figure

confirms the same pattern already observed in the pricing table: for each strike, the Tsitsiklis–Van Roy style method produces a higher price estimate than LSMC. This makes the method difference easier to see at a glance and reinforces the conclusion that the two regression-based pricing procedures are qualitatively similar but quantitatively distinct.

The bar chart also shows that the difference between the two methods is not uniform across contracts. The gap is largest for the lowest strike and becomes smaller as the strike increases. Although the experiment is limited to one underlying asset and one maturity, this pattern suggests that approximation differences may depend on the location of the contract relative to the stopping region. In particular, when the option is less deeply in the money, the continuation-value approximation may be more sensitive to the regression architecture used by the pricing method.

The visual comparison is useful because it highlights the economic interpretation of the numerical results. LSMC appears to produce more conservative American put values, while the Tsitsiklis–Van Roy style comparator implies a larger early-exercise premium under the same market inputs. This does not by itself prove that one method is universally more accurate than the other, but it shows that methodological design choices can materially affect price estimates even in a relatively simple simulation environment.



The historical AAPL price plot provides additional context for the experiment. It shows that the underlying asset has experienced substantial variation over the sample period, which helps justify the use of a simulation-based pricing framework rather than relying only on a static

single-point description of the market environment. At the same time, the historical-price plot should be interpreted only as background context: in the pricing model itself, the historical sample is summarized through the current spot price and the estimated historical volatility rather than being used directly as future price paths.

Taken together, the table and figures show that the project results are internally coherent. The tabulated outputs establish the numerical ordering of prices, the exercise-boundary figure provides intuition about the stopping behavior implied by LSMC, and the method-comparison chart illustrates the systematic valuation gap between the two regression-based approaches.

6 Critical evaluation

6.1 Strengths of LSMC / regression simulation methods

- Scales better than trees/PDE in high dimensions.
- Flexible: supports path-dependent state variables and custom payoff logic.
- Natural fit for enterprise Monte Carlo risk stacks.
- Easy to parallelize.

6.2 Weaknesses / limitations

- **Basis risk / approximation error:** poor basis choice biases continuation estimates.
- **Look-ahead and finite-sample bias** risk if implementation is careless.
- **Exercise boundary instability** in sparse ITM regions.
- **Model risk:** under realistic markets, GBM with constant volatility/rates is often insufficient.
- **Tail events** underrepresented in standard MC unless variance reduction / stress design is used.

6.3 Does this address real-world needs appropriately?

Partly yes. LSMC is practically valuable and remains widely used. But institutions using it naively (simple GBM, fixed polynomial basis, weak diagnostics) may systematically misprice early-exercise features and risk exposures, especially in stressed regimes or products with complex state dependence.

6.4 Where institutions can run into trouble

- Underfitting continuation values for deep OTM/ITM states.
- Overfitting with high-order basis and limited effective sample.
- Unstable Greeks due to regression noise.
- Governance gaps: insufficient backtesting against benchmarks and market quotes.

6.5 Suggestions for improvement

1. **Richer dynamics:** local/stochastic volatility, stochastic rates, jumps when needed.
2. **Regularized / adaptive regression:** orthogonalized basis, cross-validation, or modern ML regressors with strong controls.
3. **Variance reduction:** antithetic/control variates, quasi-MC.
4. **Dual bounds** (e.g., Andersen-Broadie style) to quantify pricing bias intervals.
5. **Robust model validation:** benchmark against lattice/PDE in low dimensions, scenario stress testing, and out-of-sample stability checks.

7 Reproducibility

This project is designed so that the main numerical results and report outputs can be reproduced from the project root directory with a small number of commands. The implementation requires a standard Python environment together with the packages `yfinance`, `pandas`, `numpy`, and `matplotlib`. In addition, Quarto is required in order to render the final report from the `.qmd` source file.

From the project root, the numerical experiment can be reproduced by first installing the required Python packages and then running the main study script:

```
python -m pip install yfinance pandas numpy matplotlib
python src/run_study.py
```

Running `run_study.py` downloads the underlying price data, estimates the model inputs, computes the European and American option prices, and regenerates the output files used in the report. In particular, it updates the pricing table in `data/pricing_results.csv` and recreates the figures stored in the `visuals` folder.

After the numerical outputs have been generated, the written report can be rendered from the Quarto source file with:

```
quarto render report/american_options_lsmc_study.qmd --to pdf
```

If PDF rendering tools are unavailable, the report can instead be rendered to HTML with:

```
quarto render report/american_options_lsmc_study.qmd --to html
```

Overall, the project is reproducible in the sense that the market inputs, numerical outputs, and written report are all generated from the same codebase and file structure. This makes it possible to rerun the experiment with updated market data or modified parameter settings while preserving a clear connection between the implementation and the final written analysis.